Museum Metadata Detective

Prioritising Data-Quality Review in Open Museum Collections

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1. Introduction

Museums increasingly publish collection metadata so that artworks can be searched, studied and reused computationally. This creates a practical data-quality problem: a catalogue may contain missing fields, inconsistent representations, implausible dates or statistically unusual values, yet checking every record manually is expensive. The MoMA snapshot used in this project contains 160,705 artwork records and 15,934 artist records. Even a very quick 30-second manual check of every artwork would require about 1,339 staff hours. At the same time, fully automatic correction is inappropriate because cultural metadata contains uncertainty, posthumous editions, organisations as creators, historical interpretation and other legitimate exceptions.

Museum Metadata Detective therefore reframes the task from automatic error correction to transparent review prioritisation. The research question is: can a small set of interpretable data-quality signals reduce a very large museum catalogue to a manageable queue of records for expert review? The project uses four signal families: completeness, representation/standardisation, logical consistency and statistical anomaly screening. The final output is a review tier and an explanation for each flag, not an automatic correction. This directly responds to the Assignment 1 feedback by making the operational value and the distinctive contribution more explicit.

The goal is useful triage under limited review capacity. A successful system should (1) identify explainable candidate records, (2) place the strongest contradictions first, (3) reduce unnecessary manual checking, and (4) expose possible bias when some departments are flagged more often because their metadata is structurally different. Better metadata supports public discovery, scholarly reuse, cross-institutional integration and downstream data-science or AI applications (Bruce & Hillmann, 2004; Bellini & Nesi, 2013).

2. Related Work

Metadata quality has long been treated as multidimensional. Bruce and Hillmann (2004) describe dimensions including completeness, accuracy, conformance and logical consistency. Bellini and Nesi (2013) operationalised completeness, accuracy and consistency for cultural-heritage repositories, showing that missing, non-standard or incorrect metadata can directly affect search and discovery. Lorenzini, Rospocher and Tonelli (2021b) likewise studied automated completeness assessment for cultural-heritage repositories.

More advanced approaches use machine learning or semantic infrastructure. Lorenzini et al. (2021a) framed textual-description quality as a classification problem over more than 100,000 records and reported F1 of about 0.85. Bobasheva, Gandon and Precioso (2022) combined symbolic reasoning and machine learning over a knowledge graph to support museum curators, while Candela (2023) assessed consistency in Linked Open Data from cultural-heritage institutions. These studies show that automation can support quality control, but they often depend on labelled text, linked-data structures, semantic reasoning or more specialised pipelines.

The contribution here is narrower and operational: a transparent triage workflow for ordinary tabular museum data when human review time is limited. Rather than claiming that rule-based metadata checking is new, the project combines several understandable signals into explicit review tiers and evaluates how sharply they reduce the first-pass review queue. This is deliberately human-in-the-loop. Recent work on responsible AI in museum collection management also stresses that automated systems should support rather than replace professional judgement, especially where metadata and classification may reflect historical bias or uncertainty (Klarzak, 2026).

3. Business Model

The project follows an information-based differentiation model: the museum uses its own data more effectively to improve an existing digital collection service. The analytics are mainly descriptive (where quality signals occur) with a prescriptive decision-support output (which records should be reviewed first). The value is operational rather than direct revenue: lower screening effort, clearer review priorities, more dependable structured data and safer reuse in research or AI.

Primary stakeholders are curators, registrars and collection-data managers. Researchers and digital teams benefit from more reliable structured data, while the public benefits indirectly from improved catalogue search and interpretation. The main business risk is false positives: unusual does not mean wrong. The project therefore keeps every rule explainable and leaves correction authority with domain experts.

4. Characterising and Analysing Data

4.1 Data sources and the 4Vs

The demonstration uses MoMA's public Artworks and Artists CSV files. The uploaded snapshot contains 160,705 artworks (30 columns; about 69.7 MB) and 15,934 artists (9 columns; about 1.0 MB). MoMA states that the research data include basic artwork and artist metadata and that some records are incomplete or not curatorially approved; the repository is revised over time (Museum of Modern Art, n.d.).

4V	Application to MoMA data
Volume	160,705 artworks and 15,934 artists; easily handled locally, but too large for record-by-record review.
Variety	Text, categorical values, IDs, dates, creator biographies and numerical dimensions; some artworks have multiple creators.
Velocity	Low rather than streaming. The collection is periodically revised, so reproducibility requires recording the dataset snapshot/version.
Veracity	Central project issue: missing fields, formatting variants, uncertain dates and unusual values can coexist with legitimate cultural exceptions.

Table 1. 4V characterisation of the demonstration data.

R, tidyverse and R Markdown are sufficient for the present CSV-scale demonstration. Raw files should be preserved unchanged, while derived tables and flags are reproducibly generated from code. If the project expanded across many institutions, a relational database such as PostgreSQL would be preferable for joins, versioned records and querying; cloud object storage and scheduled pipelines could support larger periodic updates. Distributed processing is not justified by the current data volume.

4.2 Overall analysis approach

The proposed production workflow has five stages. First, ingest and version the raw catalogue. Second, standardise fields and profile missingness. Third, calculate independent quality signals: missing core fields; representation problems such as unwanted whitespace; deterministic temporal rules; and robust numerical anomaly rules. Fourth, convert signals into review tiers. Logical contradictions are Tier 1 because they directly violate a chronological constraint; records with two or more independent signals are Tier 2; single-signal records are Tier 3. Fifth, experts review the queue and record outcomes.

A full deployment should evaluate precision at the top of the queue (for example, the proportion of the first 50 or 100 records that experts judge to need correction or clarification), review-reduction ratio, and false-positive rates by department. These measures are more appropriate than a generic prediction-accuracy score because the practical objective is efficient triage, not autonomous historical judgement. If enough reviewed examples

accumulate later, supervised or anomaly-detection models could be added, but the first version intentionally tests the value of simpler methods.

4.3 R demonstration and feasibility results

The R demonstration applies four deliberately simple signals. Completeness checks four core fields: Artist, Date, Medium and Dimensions. Representation screening checks unwanted leading/trailing whitespace in Medium. Logical screening is conservative: only single-creator records with a simple year expression are tested for an artwork year earlier than the creator's recorded BeginDate. Statistical screening flags height or width above the 99.9th percentile. Extreme dimensions are treated as unusual, not erroneous.

Signal	Records	Share of artworks
At least one missing core field	17,170	10.68%
Medium formatting flag	1,243	0.77%
Logical artwork-before-creator flag	10	0.006%
Extreme height/width flag	254	0.16%

Table 2. Main demonstration signals.

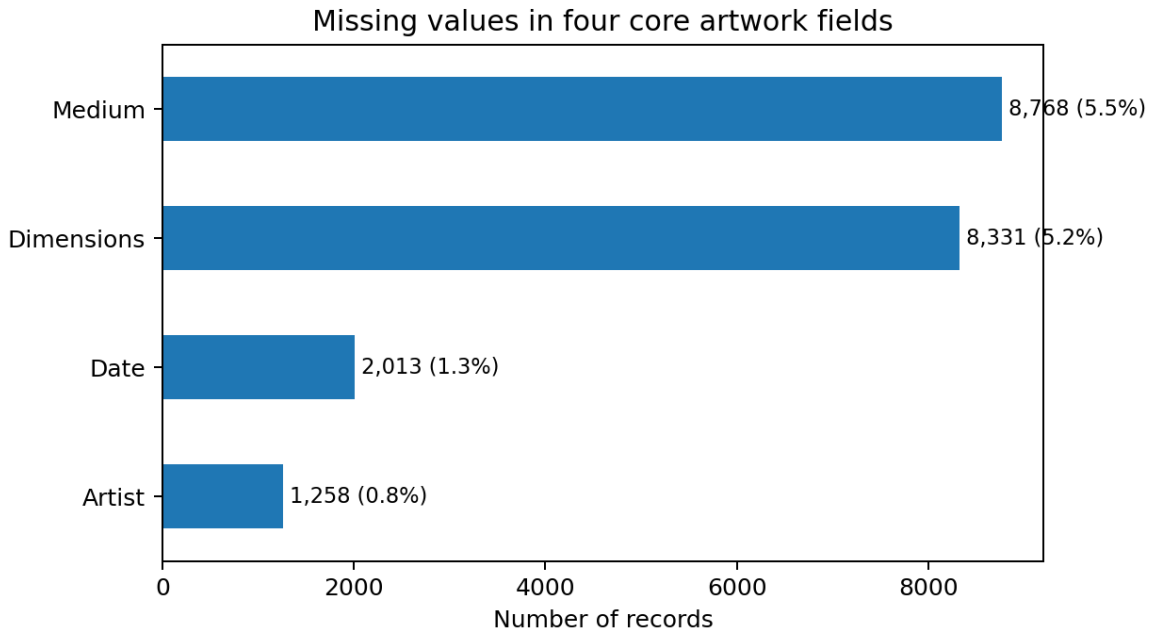


Figure 1. Missing values in the four core artwork fields.

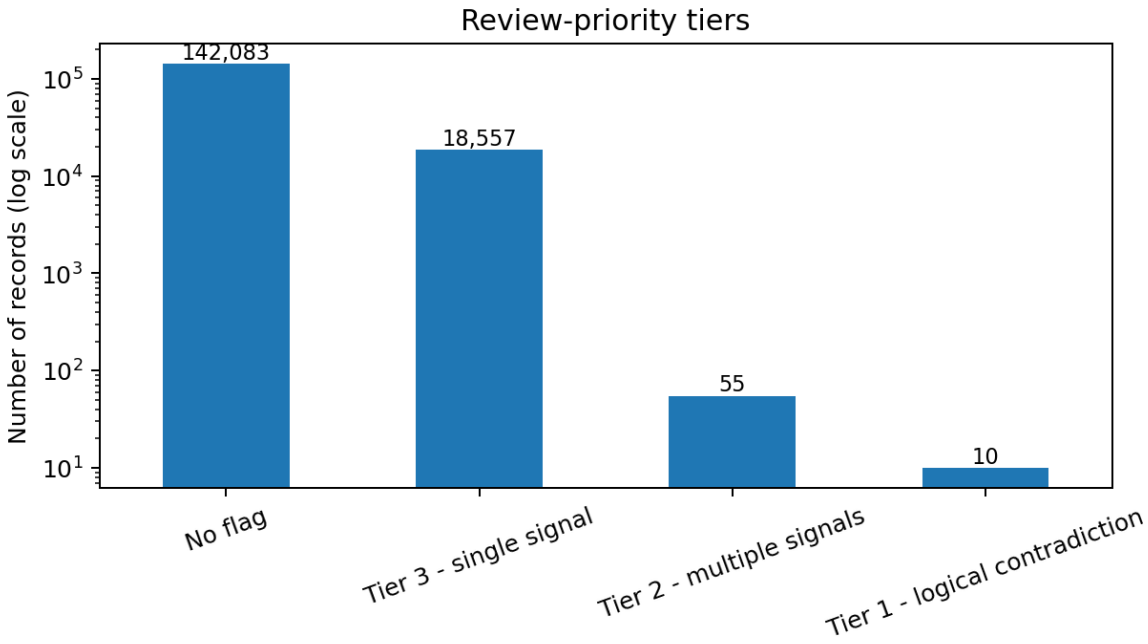


Figure 2. Review-priority tiers generated by the demonstration.

The four core fields are complete for 143,535 artworks; 17,170 records have at least one missing core field. Medium contains 1,243 records with leading or trailing whitespace. Simply trimming this whitespace reduces distinct Medium strings from 23,198 to 22,951, showing that a small standardisation step removes 247 spurious variants. The robust size rule flags 254 records, many of which are genuinely large installations or works on paper; this is useful evidence that statistical rarity must not be equated with error.

The priority system leaves 142,083 records (88.41%) with no flag, 18,557 (11.55%) with a single lower-priority signal, 55 with multiple signals, and 10 Tier 1 logical contradictions. Therefore the first-pass high-priority queue is only 65 records, about 0.04% of the catalogue. One Tier 1 result is the Assignment 1 example: Holländische Kunstausstellung (1903) is attributed to Johan Thorn Prikker while the artist table records BeginDate 1932. The rule also flags organisations such as design studios whose recorded BeginDate may represent a founding year rather than a person's birth. This demonstrates both usefulness and the need for expert interpretation.

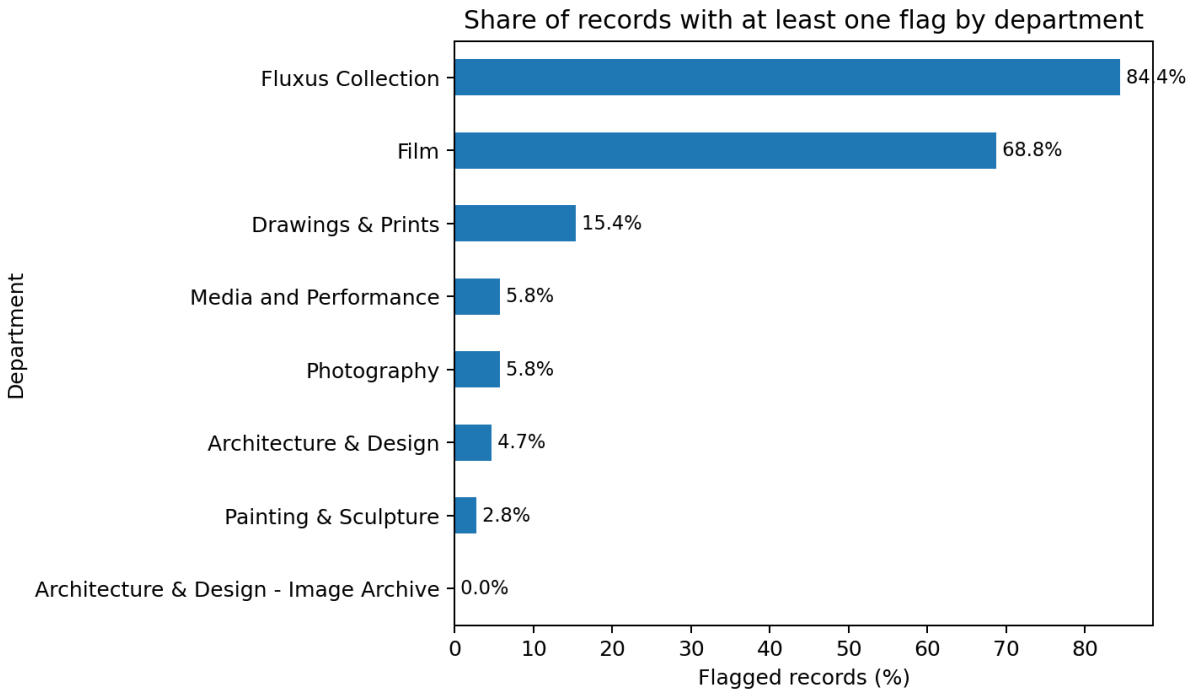


Figure 3. Flag rates vary strongly by department.

Flagging is not evenly distributed. In this snapshot, about 84.4% of Fluxus Collection records and 68.8% of Film records receive at least one signal, compared with 4.7% in Architecture & Design and 2.8% in Painting & Sculpture. This does not prove that Fluxus or Film metadata are worse. Their record structures may simply have different expectations for creator, date, medium or dimensions. The department comparison therefore becomes a governance check against blindly treating missingness as error.

5. Standards, Data Governance and Management

The project follows the metadata-quality dimensions used by Bruce and Hillmann (2004), particularly completeness, conformance and logical consistency. The data-science process also follows basic reproducibility principles: preserve raw source files, document the snapshot date, perform transformations in code, retain ObjectID as the record key, and separate original data from derived flags. MoMA publishes the research dataset openly, so accessibility is high; however, source attribution and licence conditions must still be preserved.

The main ethical issue is not privacy but interpretive bias. Older, under-documented or structurally different collections may be flagged more often even when their records are legitimate. The demonstration already shows large department-level differences. For this reason, the system must never auto-correct authoritative museum metadata, should show the exact reason for each flag, and should monitor false-positive rates across departments and collection types. Human review remains the final authority.

References

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